

SITE / FACILITY INFORMATION			
FACILITY:	Cadillac WWTP	Digestion Method: 1 (Anaerobic = 1, Aerobic = 2, Lime stabilized = 3) App. Method: Injection (Inj, Inc, Sfc, not case sensitive) Org. N-1 yr. ago: Org. N-2 yrs. ago: TOTAL ACRES: 55 ACRES USED: 55 App. Rate (Gal/Ac) 6,447 App. Rate (DT/Ac) 0.88 Org. N App. (#/ac) 94	
MONTH & YEAR:	September 26, 2016		
FACILITY NPDES #	MI0020257		
MDEQ SITE ID #:	21N09W30-GR01		
COUNTY:	Wexford		
TOWNSHIP:	Clam Lake		
SECTION:	30		
OWNER:	Gordon Roush		
FARMER:	John Rydquist		
ADDRESS:	11600 E. 50 Rd		
ADDRESS:	Cadillac, MI 49601		
TELEPHONE:	231-884-0719		
SUPERINTENDANT	Jeff Dietlin		
LATITUDE:	N44* 18'33"		
LONGITUDE:	W85* 43'82"		

Seasons Used:	2	Field Complete?	Yes
		Crop Year?	2016
		Overlap?	No

SOIL TEST RESULTS			
SAMPLE DATE:	07/23/15	NEXT CROP:	Corn
CEC:	4.4	CROP YLD GOAL:	150
PH:	6.5	NIT. REC. (lbs/ac):	175
LIME INDEX:	71.0	ADJ. N REC.	175
BRAY P1 (ppm):	25	AVAIL. N APP.	84
POTASSIUM (ppm):	74	SOIL & APP. P	129
CALCIUM (ppm):	692	P ₂ O ₅ REC. (lbs/ac):	55
MAGNESIUM (ppm):	88	K ₂ O REC. (lbs/ac):	55
		LIME REC. (T/ac):	0.0

Accumulated History		
ARSENIC (As)	0.05	lbs/acre
CADMIUM (Cd)	0	lbs/acre
CHROMIUM (Cr)	0.26	lbs/acre
COPPER (Cu)	2.14	lbs/acre
LEAD(Pb)	0.09	lbs/acre
MERCURY (Hg)	0	lbs/acre
MOLYB. (Mo)	0.05	lbs/acre
NICKEL (Ni)	0.07	lbs/acre
SELENIUM (Se)	0.01	lbs/acre

[illegible]

ZINC (Zn)	3.45	lbs/acre
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	BIOSOLIDS ANALYSIS	
Lab ID	KAR	612234
Sample Name		Cadillac WWTP
Date Collected		9/12/2016
Matrix		Sludge
Density	lbs/gal	8.45
Mercury	mg/kg	0.56
Nitrogen,Ammonia	mg/kg	37,000.00
Nitrogen,Total	mg/kg	0.00
Nitrogen,Total Avail.	lb/ton	0.00
Nit.,Total Kjeldahl	mg/kg	90,200.00
pH	STD Units	7.70
Total Phosphorus	mg/kg	45,000.00
Nitrogen,Nitrate	mg/kg	3.10
Cadmium	mg/kg	3.80
Calcium	mg/kg	16,000.00
Chromium	mg/kg	50.50
Copper	mg/kg	467.00
Lead	mg/kg	39.00
Magnesium	mg/kg	2,740.00
Molybdenum	mg/kg	20.20
Nickel	mg/kg	15.00
Potassium	mg/kg	1,740.00
Zinc	mg/kg	828.00
Arsenic	mg/kg	15.50
Selenium	mg/kg	3.80
Total Solids	%	3.24
Total Volatile Solids	%	53.70

BIOSOLIDS APPLIED

[illegible]

Source:	Cadillac WWTP	Owner:	Gordon Roush	App. Rate (Gal/Ac)	6,447
Date:	09/26/16	Farmer:	John Rydquist	App. Rate (DT/Ac)	0.88
NPDES #:	MI0020257	Address:	11600 E. 50 Rd	Useable Acres	55.0
MDEQ Site #:	21N09W30-GR01	City:	Cadillac, MI 49601	Acres Applied	55.0
County:	Wexford	Telephone:	231-884-0719	Seasons Used	2
Township:	Clam Lake	Latitude:	N44° 18'33"		
Section:	30	Longitude:	W85° 43'82"		

BIOSOLIDS ADDITIONS									
<u>Nutrient</u>	<u>Biosolids Additions</u>		<u>Residual Organic N Credit</u>		<u>Soil Fertility Test</u>		<u>Total Estimated Nutrients</u>		
Nitrogen (lb N/Ac)	84	+	0	+	0	lb N	=	84	lb N/Ac
Phos (lb P/Ac)	79			+	50	lb P	=	129	lb P/Ac
Pot (lb K/Ac)	3			+	148	lb K	=	151	lb K/Ac

SOIL ANALYSIS AND CROP INFORMATION					
C.E.C.:	4.4	P (lbs/acre):	50	K (lbs/acre):	148
Soil pH:	6.5	P (ppm):	25	K (ppm):	74
Lime Index:	71.0	Ca (lbs/acre):	1384	Mg (lbs/acre):	176

FERTILIZER RECOMMENDATIONS (lbs./a)						
Crop to be Fertilized	Yield Goal	N	P2O5	K2O	Lime	
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Corn	150	175	55	55	0.0	Lbs./A

BIOSOLIDS ANALYSES									
Date: 12-Sep		Dry wt. basis	Lbs/dry ton	Lbs Applied Per Acre	Allowable Lifetime Lbs/acre	Previous Applied lb/acre	Total Applied lb/acre	Life in Years	% Cum.
		-----	-----	-----	-----	-----	-----	-----	-----
Density	(mg/l)-->	1.01	----	----					
Weight	(Lb/Gal)>	8.45	----	----					
Solids	(%)----->	3.24	----	----					
TKN	(%)----->	9.02	----	----					
Amm. N	(%)----->	3.70	----	----					
Nit. N	(%)----->	0.000	----	----					
Total Plant Avail. N----->			95	84					
Total P	(%)--->	4.50	90	79					
Total K	(%)--->	0.17	3	3					
Total Ca	(%)--->	1.60	32.00	28.24					
Total Mg	(%)--->	0.27	5.48	4.84					
Total As	(ppm)->	15.50	0.031	0.027	37	0.05	0.08	1,352	0.21%
Total Cd	(ppm)->	3.80	0.008	0.007	35	0.00	0.01	5,219	0.02%
Total Cr	(ppm)->	50.50	0.101	0.089	N/A	0.26	0.35	N/A	N/A
Total Cu	(ppm)->	467.00	0.934	0.824	1,335	2.14	2.96	1,620	0.22%
Total Pb	(ppm)->	39.00	0.078	0.069	267	0.09	0.16	3,879	0.06%
Total Hg	(ppm)->	0.56	0.001	0.001	15	0.00	0.00	15,176	0.01%
Total Mo	(ppm)->	20.20	0.040	0.036	N/A	0.05	0.09	N/A	N/A
Total Ni	(ppm)->	15.00	0.030	0.026	274	0.07	0.10	10,350	0.04%
Total Se	(ppm)->	3.80	0.008	0.007	89	0.01	0.02	13,270	0.02%
Total Zn	(ppm)->	828.00	1.656	1.461	2,492	3.45	4.91	1,705	0.20%

Month/Year: September-16																Michigan Department of Environmental Quality														MDEQ Site # 21N09W30-GR01				CROP & SOIL DATA	
Plant ID: MI0020257																Water Division														# of seasons utilized: 2				Crop to be fertilized: Corn	
Superintendent: Jeff Dietlin																Biosolids Recycling Tracking Sheet														Acres utilized this month: 55					
Facility Name: Cadillac WWTP		Use of this form is voluntary. Submission of this form is not required																												Total acres in site: 55				Subsequent Crop:	
DAY	BIOSOLIDS APPLIED				NUTRIENTS								METALS																						
	MG	Solids %	Vol. %	Dry Tons	TKN %	NH4 %	NO3 %	TKN lb/ac	Phosphorus % lb/ac	Potassium % lb/ac	Arsenic mg/kg lb/ac	Cadmium mg/kg lb/ac	Copper mg/kg lb/ac	Lead mg/kg lb/ac	Mercury mg/kg lb/ac	Molybdenum mg/kg lb/ac	Nickel mg/kg lb/ac	Selenium mg/kg lb/ac	Zinc mg/kg lb/ac																
1	0.107	3.2		14.63792	9.02	3.70	0.00031		4.50		0.17		15.5		3.8		467		39.0		0.6		20.2		15.0		3.8		828		Bray P1 (ppm): 25				
2	0.079	3.2		10.78584	9.02	3.70	0.00031		4.50		0.17		15.5		3.8		467		39.0		0.6		20.2		15.0		3.8		828		K (ppm): 74				
3	0.017	3.2		2.311251	9.02	3.70	0.00031		4.50		0.17		15.5		3.8		467		39.0		0.6		20.2		15.0		3.8		828		Crop Yield Goal: 150				
4	0.084	3.2		11.55625	9.02	3.70	0.00031		4.50		0.17		15.5		3.8		467		39.0		0.6		20.2		15.0		3.8		828		Nitrogen Recom. (lb/ac) 175				
5	0.068	3.2		9.245003	9.02	3.70	0.00031		4.50		0.17		15.5		3.8		467		39.0		0.6		20.2		15.0		3.8		828						
6	0.000	0.0		0	9.02	3.70	0.00031		4.50		0.17		15.5		3.8		467		39.0		0.6		20.2		15.0		3.8		828						
7	0.000	0.0		0	9.02	3.70	0.00031		4.50		0.17		15.5		3.8		467		39.0		0.6		20.2		15.0		3.8		828						
8	0.000	0.0		0	9.02	3.70	0.00031		4.50		0.17		15.5		3.8		467		39.0		0.6		20.2		15.0		3.8		828						
9	0.000	0.0		0	9.02	3.70	0.00031		4.50		0.17		15.5		3.8		467		39.0		0.6		20.2		15.0		3.8		828						
10	0.000	0.0		0	9.02	3.70	0.00031		4.50		0.17		15.5		3.8		467		39.0		0.6		20.2		15.0		3.8		828						
11	0.000	0.0		0	9.02	3.70	0.00031		4.50		0.17		15.5		3.8		467		39.0		0.6		20.2		15.0		3.8		828						
12	0.000	0.0		0	9.02	3.70	0.00031		4.50		0.17		15.5		3.8		467		39.0		0.6		20.2		15.0		3.8		828						
13	0.000	0.0		0	9.02	3.70	0.00031		4.50		0.17		15.5		3.8		467		39.0		0.6		20.2		15.0		3.8		828						
14	0.000	0.0		0	9.02	3.70	0.00031		4.50		0.17		15.5		3.8		467		39.0		0.6		20.2		15.0		3.8		828						
15	0.000	0.0		0	9.02	3.70	0.00031		4.50		0.17		15.5		3.8		467		39.0		0.6		20.2		15.0		3.8		828						
16	0.000	0.0		0	9.02	3.70	0.00031		4.50		0.17		15.5		3.8		467		39.0		0.6		20.2		15.0		3.8		828						
17	0.000	0.0		0	9.02	3.70	0.00031		4.50		0.17		15.5		3.8		467		39.0		0.6		20.2		15.0		3.8		828						
18	0.000	0.0		0	9.02	3.70	0.00031		4.50		0.17		15.5		3.8		467		39.0		0.6		20.2		15.0		3.8		828						
19	0.000	0.0		0	9.02	3.70	0.00031		4.50		0.17		15.5		3.8		467		39.0		0.6		20.2		15.0		3.8		828						
20	0.000	0.0		0	9.02	3.70	0.00031		4.50		0.17		15.5		3.8		467		39.0		0.6		20.2		15.0		3.8		828						
21	0.000	0.0		0	9.02	3.70	0.00031		4.50		0.17		15.5		3.8		467		39.0		0.6		20.2		15.0		3.8		828						
22	0.000	0.0		0	9.02	3.70	0.00031		4.50		0.17		15.5		3.8		467		39.0		0.6		20.2		15.0		3.8		828						
23	0.000	0.0		0	9.02	3.70	0.00031		4.50		0.17		15.5		3.8		467		39.0		0.6		20.2		15.0		3.8		828						
24	0.000	0.0		0	9.02	3.70	0.00031		4.50		0.17		15.5		3.8		467		39.0		0.6		20.2		15.0		3.8		828						
25	0.000	0.0		0	9.02	3.70	0.00031		4.50		0.17		15.5		3.8		467		39.0		0.6		20.2		15.0		3.8		828						
26	0.000	0.0		0	9.02	3.70	0.00031		4.50		0.17		15.5		3.8		467		39.0		0.6		20.2		15.0		3.8		828						
27	0.000	0.0		0	9.02	3.70	0.00031		4.50		0.17		15.5		3.8		467		39.0		0.6		20.2		15.0		3.8		828						
28	0.000	0.0		0	9.02	3.70	0.00031		4.50		0.17		15.5		3.8		467		39.0		0.6		20.2		15.0		3.8		828						
29	0.000	0.0		0	9.02	3.70	0.00031		4.50		0.17		15.5		3.8		467		39.0		0.6		20.2		15.0		3.8		828						
30	0.000	0.0		0	9.02	3.70	0.00031		4.50		0.17		15.5		3.8		467		39.0		0.6		20.2		15.0		3.8		828						
31	0.000	0.0		0	9.02	3.70	0.00031		4.50		0.17		15.5		3.8		467		39.0		0.6		20.2		15.0		3.8		828						
Average		3.24			9.02	3.70	0.00031		4.50		0.17		15.50		3.8		467		39.0		0.56		20.2		15.0		3.80		828						
Totals	This Mo	0.355		48.5				84		79.4		3.1		0.0		0.0		0.8		0.1		0.0		0.0		0.0		0.0		1.5	BIOSOLIDS ANALYSIS DATE: 9/12/2016				
	This Yr																														SOIL ANALYSIS DATE: 7/23/2015				
	Cumulative													0.08		0.01		2.96		0.16		0.00		0.09		0.10		0.02		4.91					

42639

Facility NPDES # MI0020257

Lat/Long: N44* 18'33"

Total acres in site: 55

K 148 lbs/a

[illegible]

[illegible]

Org. N-1 yr. Ago (lbs/ac):	0	0	Residual N Credit
Org. N-2 yrs. Ago (lbs/ac):	0	0	Residual N Credit

4368.26	168.906	1.50462	0.368876	4.902163	45.33287	3.785829	0.054361	1.960865
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As	37
Cd	4.5
Cu	110
Pb	268
Hg	15
Ni	44
Se	89
Zn	220

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15.00		3.80		828.00		1.60		0.27		Err:509
15.00		3.80		828.00		1.60		0.27		Err:509
15.00		3.80		828.00		1.60		0.27		Err:509
15.00		3.80		828.00		1.60		0.27		Err:509
15.00		3.80		828.00		1.60		0.27		Err:509
15.00		3.80		828.00		1.60		0.27		Err:509
15.00		3.80		828.00		1.60		0.27		Err:509
15.00		3.80		828.00		1.60		0.27		Err:509
15.00		3.80		828.00		1.60		0.27		Err:509
15.00		3.80		828.00		1.60		0.27		Err:509
15.00		3.80		828.00		1.60		0.27		Err:509
15.00		3.80		828.00		1.60		0.27		Err:509
15.00		3.80		828.00		1.60		0.27		Err:509
15.00		3.80		828.00		1.60		0.27		Err:509
15.00		3.80		828.00		1.60		0.27		Err:509
15.00		3.80		828.00		1.60		0.27		Err:509
15.00		3.80		828.00		1.60		0.27		Err:509
0.026		0.007		1.461		28.239		4.836		Err:509
0.026		0.007		1.461		28.239		4.836		Err:509
0.030		0.008		1.637		31.628		5.416		Err:509
0.096		0.017		4.911		28.239		4.836		Err:509
0.108		0.019		5.501		31.628		5.416		Err:509

0.07	0.01	3.45	0	0	0
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ni	se	zn	ca	mg	so4
0.439138	0.111248	24.2404	468.413487	80.21581	Err:509
0.323575	0.081972	17.86135	345.14678	59.10639	Err:509
0.069338	0.017566	3.827431	73.9600243	12.66565	Err:509
0.346688	0.087828	19.13716	369.800122	63.32827	Err:509
0.27735	0.070262	15.30973	295.840097	50.66262	Err:509
0	0	0	0	0	Err:509
0	0	0	0	0	Err:509
0	0	0	0	0	Err:509
0	0	0	0	0	Err:509
0	0	0	0	0	Err:509

[illegible]

Application Date	Facility	Facility #	Site I D	County	Township	Section
###	Cadillac WWTP	MI0020257	21N09W30-GR01	Wexford	Clam Lake	30

Owner	Farmer	Farmer Street Ad	Farm City, St.	Zip	Frm Telephone	Entity Super	Latitutde
Gordon Roush	John Rydquist	11600 E. 50 Rd	Cadillac, MI	49601	231-884-0719	Jeff Dietlin	N44* 18'33"

Longitude	Season Used	Dig Method	App Method	Ttl Acres	Acres Used	Soil Date
W85* 43'82"	2	1	Injection	55	55	7/23/2015

CEC	Ph	Lime Index	Bray P1	Pot	Calcium	Magnesium
4.4	6.5	71	25	74	692	88

Intend Crop	Yld Goal	N Rec	Phos Rec	Pot Rec	Lime Rec
Corn	150	175	55	55	0

Facility	Facility #	Lab I.D.	Lab Ref #	Sample ID	Date Collected	Matrix	Density
Cadillac WWT	MI0020257	KAR	612234	Cadillac WW	9/12/2016	Sludge	8.45

Hg	N Amonia	N Total	N Ttl.AvaIL #/T	N Ttl Kjeldahl	pH	Ttl. Phos	N Nitrate
0.56	37000	0	0	90200	7.7	45000	3.1

Cd	Ca	Cr	Cu	Pb	Mag	Mo	Ni
3.8	16000	50.5	467	39	2740	20.2	15

K	Zn	As	Se	Ttl. Solids	Ttl. Vol. Solids
1740	828	15.5	3.8	3.24	53.7

Facility	Facility #	Site ID	Date 1	Gallons 1	% Solids 1	Date 2	Gallons 2
adillac WW	MI0020257	IN09W30-GR	10/4/2016	106,932	3.24	10/5/2016	78,792

% Solids 2	Date 3	Gallons 3	% Solids 3	Date 4	Gallons 4	% Solids 4	Date 5
3.24	10/6/2016	16,884	3.24	10/7/2016	84,420	3.24	10/10/2016

Gallons 5	% Solids 5	Date 6	Gallons 6	% Solids 6	Date 7	Gallons 7	% Solids 7
67,536	3.24						

Date 8	Gallons 8	% Solids 8	Date 9	Gallons 9	% Solids 9	Date 10	Gallons 10
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% Solids 10	Date 11	Gallons 11	% Solids 11	Date 12	Gallons 12	% Solids 12	Date 13
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Gallons 13	% Solids 13	Date 14	Gallons 14	% Solids 14	Date 15	Gallons 15	% Solids 15
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Date 16	Gallons 16	% Solids 16	Date 17	Gallons 17	% Solids 17	Date 18	Gallons 18
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% Solids 18	Date 19	Gallons 19	% Solids 19	Date 20	Gallons 20	% Solids 20	Date 21
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Gallons 21	% Solids 21	Date 22	Gallons 22	% Solids 22	Date 23	Gallons 23	% Solids 23
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Date 24	Gallons 24	% Solids 24	Date 25	Gallons 25	% Solids 25	Date 26	Gallons 26
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% Solids 26	Date 27	Gallons 27	% Solids 27	Date 28	Gallons 28	% Solids 28	Date 29
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Gallons 29	% Solids 29	Date 30	Gallons 30	% Solids 30	Date 31	Gallons 31	% Solids 31
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